

Agentic-SAM v2

An AI Assistant for Ultrasound-Guided Radiofrequency Ablation of PTMC

User Manual

*Real-time PTMC segmentation and tracking · ablation-coverage monitoring ·
conversational surgical assistance*

Intended use. This software is a research and workflow-prototyping tool. It provides *decision support only*. It does not diagnose, does not measure thermal ablation, and must never be used as the sole basis for an intraoperative decision. Every overlay, measurement, and report must be verified visually by the operating surgeon.

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1 What This Application Does

Agentic-SAM v2 is an AI assistant that watches the intraoperative ultrasound stream alongside the surgeon during radiofrequency ablation (RFA) of a papillary thyroid microcarcinoma (PTMC). The surgeon marks the lesion once; from that moment the system segments and follows it frame by frame, measures how the visible target changes, keeps a timestamped record of the procedure, and answers questions in natural language — all on local hardware, with no patient data leaving the machine.

1.1 Feature Summary

Feature	What it does
Click-to-segment targeting	The surgeon pauses the video on any frame and clicks the PTMC. MedSAM2 segments it on that frame immediately.
Continuous lesion tracking	Playback resumes with the segmentation overlaid, updated on every subsequent frame, so the lesion stays outlined while the probe and anatomy move.
Live AI monitor	Every few seconds a short situational note appears: tracking stability, size changes, and what may be worth re-checking visually.
Ablation-coverage monitoring	The first segmented frame is stored as the baseline lesion. Coverage and residual percentages are computed continuously as ablation proceeds.
Size-drop alerts	When the visible target shrinks substantially relative to the session maximum, an alert is written to the procedure log with an estimated percentage.
Conversational assistance	Questions can be asked at any time, about the current frame or the procedure so far. Tracking pauses while the assistant answers, then resumes.
Text-prompted segmentation	Structures can also be segmented from a phrase such as <code>thyroid nodule</code> or <code>ablation zone</code> , without drawing a box.
Procedure event log	Phase changes, tracking events, alerts, snapshots, and questions are recorded with timestamps and the RFA phase in force.
Ablation report	One button produces an illustrated report: baseline lesion, final ablation pattern, coverage over time, residual assessment, limitations.
Session summary note	A separate button drafts a structured technical summary of the whole session from the event log.

1.2 How an RFA Surgeon Benefits

The target stays outlined while you work. PTMC lesions are small and low-contrast, and during ablation the picture degrades: gas and hyperechoic change obscure margins, and probe pressure shifts the plane. Marking the lesion once and having it outlined continuously reduces the effort of repeatedly re-identifying it on a changing image.

Coverage becomes visible instead of remembered. Judging which parts of a lesion have already been treated is normally done from memory of where the electrode has been. The system accumulates a *coverage map* across the whole session and reports the percentage of the baseline lesion that appears treated and the percentage that does not — turning a mental tally into an explicit, reviewable picture.

Residual regions are pointed out, not just counted. If part of the baseline lesion never shows the change associated with treatment, it is highlighted in red on the ablation map and its location is described in the report — a prompt to visually re-check that region before finishing.

An assistant that answers rather than distracts. Questions such as “how much is ablated?” or “did I cover the whole lesion?” are answered from measured data in one short reply. Answers are deliberately terse: an operating room has no time for prose.

Documentation writes itself. Every notable event is timestamped during the case. The post-procedure report is generated from that record, so producing documentation costs one button press rather than recall after the fact.

Everything runs locally. Segmentation models and the language model both run on the workstation’s own GPU. No images or clinical text are transmitted off the machine.

How to read “coverage”. Coverage is a *tracking-based proxy*. The software counts a region as treated when the tracker stops capturing it, because ablated tissue becomes hyperechoic and loses its previous appearance. That signal correlates with treatment but is not a measurement of thermal dose, and it can also be produced by probe motion, plane change, or tracking error. Treat coverage figures as a prompt to look, never as proof of treatment.

2 System Overview

The application runs as a *single process* that serves the web interface and holds the segmentation models in memory, plus one local inference server for the language model.

Component	Role
MedSAM2	Segments the lesion from the surgeon’s box and tracks it through subsequent frames. This is the model behind every overlay on the Live Procedure page.
Medical-SAM3	Segments structures from a text prompt on medical ultrasound.
SAM 3.1	General-purpose text-prompted segmentation, used as a fallback when Medical-SAM3 returns nothing.
Language model (vLLM)	Runs locally through an OpenAI-compatible interface. Powers the live monitor notes, the conversational answers, and both reports.
Agent (LangGraph)	Decides which tool to use for a question, remembers the conversation, and grounds safety-critical answers in measured data.

All models are loaded *before* the interface becomes reachable, so the first segmentation of a case is as fast as every later one — there is no cold start in the middle of a procedure.

2.1 Starting the Application

```
cd /home/ubuntu/research/agentic-sam-v2
bash run.sh # uses the settings in the CONFIGURATION block
bash run.sh 2 # ... but runs everything on GPU 2
```

Startup prints its progress and takes roughly a minute (longer the first time, when the language model is downloaded):

```
[serve] LLM ready: vLLM serving Qwen/Qwen2.5-VL-7B-Instruct
[serve] MedSAM2 loaded in 3.8s
[serve] Medical-SAM3 loaded in 17.2s
[serve] SAM 3.1 loaded in 8.5s
[serve] all models warm --starting UI on 0.0.0.0:7870
```

The interface is then available at <http://<workstation>:7870>. Press **Ctrl+C** in the terminal to stop the application and the inference server together.

2.2 Configuration

The top of `run.sh` contains a configuration block; edit it rather than passing environment variables.

Setting	Default	Meaning
GPU	0	GPU the whole stack uses; the first command-line argument overrides it.
VLLM_MEM_FRACTION	0.35	Share of that GPU reserved for the language model; the segmentation models use the remainder.
APP_ENV	agentic env	Python environment running the interface and segmentation models.
VLLM_ENV	vllm_m3 env	Python environment running the inference server.
LLM_MODEL	Qwen2.5-VL-7B	Language model served locally. Must support images and tool calling.
UI_PORT	7870	Port the interface listens on.

Two further settings are worth knowing, set in the `.env` file:

- `ASAM_PIXEL_SPACING_MM` — millimetres per pixel for the ultrasound preset in use. When set, all measurements are additionally reported in mm and mm²; when left at 0, measurements are in pixels only.
- `ASAM_ABLATION_SAMPLE_SECONDS` — how often the coverage monitor samples the tracked lesion (default: every 5 seconds).

3 The Live Procedure Page

This is the page used during the case. It shows the ultrasound frame with the segmentation overlay, the live assistant, the coverage metrics, and the recent-events list.

3.1 Step-by-Step Workflow

1. **Load the video.** Choose **Sample RFA video** or **Upload** and select a file. Frames are extracted once; the frame count and rate are reported underneath.
2. **Set the RFA phase.** The **RFA phase** selector offers *Pre-op review*, *Targeting*, *Ablation*, and *Post-ablation assessment*. The current phase is stamped on every event recorded from then on, and is given to the assistant as context.
3. **Find the lesion.** Press ► **Play** to run the video, and || **Pause** when the PTMC is well seen. The **Sampled frame** slider can also be dragged to any frame; dragging it pauses playback automatically.
4. **Mark the PTMC.** Click on the lesion in the paused frame. A marker and a square target box appear. Adjust **Tracking box size (px)** beforehand if the default box does not enclose the lesion, and click again to reposition.
5. **Start tracking.** Press **Track PTMC**. The lesion is segmented on the marked frame and playback resumes automatically, with the overlay redrawn on each new frame as it is segmented. The manual marker disappears — the live overlay replaces it.
6. **Ablate.** Perform the ablation while the overlay follows the lesion. Coverage and residual percentages update as treatment proceeds, and the live assistant comments on what it observes.
7. **Pause whenever needed.** || **Pause** halts tracking; ► **Continue** resumes it from where it stopped. While paused, clicking the image again re-marks the lesion, and pressing **Track PTMC** restarts tracking from the corrected position — useful after a large probe movement.
8. **Capture anything notable.** **Snapshot to event log** stores the current overlaid frame together with the coverage figures.

Choosing where to start tracking. Start from a frame where the lesion margin is clearly visible. Tracking quality throughout the case depends heavily on the quality of that first segmentation — it also becomes the baseline against which all coverage is measured.

3.2 Reading the Right-Hand Panel

Live assistant. A short note refreshed every few seconds: a one-line status followed by at most three observations. It reports tracking stability and apparent size changes, and suggests visual checks. It never gives instructions.

Tracked PTMC area. The current segmented area in pixels, with its change relative to the largest area seen during the session.

Ablation coverage and residual PTMC. The percentage of the baseline lesion that appears treated, and the percentage that does not. Both are marked with an asterisk as proxies. The caption shows how many samples have been taken.

Recent events. The last few entries of the procedure log, each with its time and type: tracking events, alerts, snapshots, questions, and phase changes.

3.3 Asking Questions During the Procedure

Below the video, **Ask during the procedure** accepts a typed question or one of three preset questions:

- *Is the PTMC tracking stable in this frame?*
- *Has the visible PTMC area changed recently?*
- *What should be visually re-checked right now?*

Asking pauses tracking; the assistant answers using the current frame and the procedure context, and the exchange is added to the event log. Press **Continue** to resume. Ablation questions — coverage, residual, how much is treated — are always answered from the measured coverage data rather than from the model’s impression of the image.

4 The Image Analysis Page

This page is for examining a single still image: a pre-operative capture, a frame of interest, or any exported ultrasound image. It is a conversation rather than a live feed.

1. Choose **Sample PTMC image** or **Upload**.
2. Optionally click two opposite corners to draw a box around the lesion. The box is shown with its coordinates; **Clear box** removes it.
3. Ask a question by typing it or selecting a preset (*Where is the PTMC in this ultrasound image?*, *Describe this ultrasound image.*, *How large is the segmented target?*).

The assistant chooses its own approach: with a box drawn and a localization question asked, it segments inside the box with MedSAM2; without a box it segments from the text of the question; for size questions it measures the most recent segmentation; for background questions it may consult the web. The overlay appears in the conversation and in **Latest segmentation overlay**, and each answer carries an expandable **Tools used** record showing exactly which tool produced the numbers.

The conversation has memory: follow-up questions such as “and how large is it?” are understood in the context of what was just discussed. The conversation is shared with the Live Procedure page, so a question asked during the case and a follow-up asked here belong to the same thread.

5 The Procedure Report Page

This page reviews the completed case and produces documentation. Everything it shows is reconstructed from the event log, so it can be opened mid-case as well.

5.1 Session Overview

Counters for events, snapshots, the final phase, and the current area relative to the session maximum; a chart of tracked lesion area over time; the full event table (time, phase, type, message); and a gallery of all snapshots.

5.2 Ablation Coverage

- **Metrics** — final coverage and residual percentages, the baseline lesion area, and the number of samples taken.
- **Baseline PTMC** — the first tracked frame with its segmentation, i.e. the lesion as it appeared before treatment.
- **Ablation pattern** — the coverage map. **Green** marks baseline regions counted as treated; **red** marks regions never observed to change, which may not have been covered.
- **Coverage chart** — how coverage accumulated over the case, revealing whether treatment progressed steadily or plateaued.

5.3 Generating Reports

Generate ablation report produces an illustrated analysis: ablation extent over time, the spatial pattern, an assessment of residual regions and where they lie, supporting signals such as echo-brightness change, and the limitations of the method. **Generate session summary note** produces a structured technical summary of the whole session from the event log. Both can be downloaded as Markdown.

Why reports are generated on demand. All monitoring during the case is ordinary image arithmetic and costs nothing in language-model terms. The language model is invoked once per report, when the button is pressed. This keeps the live workflow fast and inference cost low.

6 The System Status Page

For checking the installation and diagnosing problems. It lists each model with its checkpoint, size, and whether it is loaded; offers manual preload buttons; shows GPU utilisation and memory; and reports the session identifier, the language-model endpoint, and the active settings including pixel spacing.

If a model shows as absent, its weights are missing — run `python downloader.py` to restore them. If the language-model status is not *ready*, the inference server did not start; check `.logs/vllm.log`.

7 Troubleshooting

Symptom	Cause and remedy
The video will not load and a Git LFS message appears	The file is a placeholder stub, not a real video. Copy the actual recording into <code>data/video/</code> or upload a different file.
Tracking stops early	The tracker reached the end of the sampled frames, or lost the target. Check the status line under the video; re-mark the lesion and press Track PTMC again.
The overlay drifts off the lesion	Probe or plane movement outstripped the tracker. Pause, click the lesion again, and restart tracking; coverage accumulated so far is retained.

Symptom	Cause and remedy
Coverage rises without ablation	The tracker is losing the target for other reasons (motion, shadowing). Coverage is a proxy; re-mark the lesion and interpret the figures with that in mind.
Live assistant shows an error	The inference server is unreachable. Check the System Status page and <code>.logs/vllm.log</code> . Tracking and segmentation continue to work without it.
<code>FileNotFoundError: 'ninja'</code> at startup	The inference environment's <code>bin</code> directory is not on <code>PATH</code> . <code>run.sh</code> handles this; ensure <code>VLLM_ENV</code> points at the environment that actually contains <code>vLLM</code> .
Out-of-memory on startup	Lower <code>VLLM_MEM_FRACTION</code> in <code>run.sh</code> , or select a GPU with more free memory.
GPU memory stays occupied after exit	Stop with Ctrl+C rather than closing the terminal, so the inference server is shut down with the application.

8 Safety and Limitations

1. **Decision support only.** The system does not diagnose and does not recommend treatment. The surgeon remains responsible for every decision.
2. **Coverage is a proxy.** It is inferred from loss of tracking against the baseline lesion, not from thermal measurement. Probe motion, plane change, shadowing, and tracking error can all produce the same signal.
3. **Measurements are pixel-based** unless a pixel spacing has been configured for the ultrasound preset in use. Millimetre values are only as accurate as that calibration.
4. **Segmentation can fail silently.** A confidently drawn outline may still be wrong. Visual verification is required at every stage.
5. **Reports are technical summaries**, generated from what the software recorded. They are not clinical operative notes and are not a substitute for them.
6. **Only what is visible is assessed.** Lesion extent outside the imaging plane, and any tissue not in view, are invisible to the system and are not represented in coverage figures.

Not a medical device. Not cleared or approved by any regulatory body. For research and workflow prototyping only.